

Question Number	Answer	Additional guidance	Mark
5(a)(i)	An answer that makes reference to the following point: chloroplast is the site of {photosynthesis / glucose production / light absorption} (1)	accept site of light {dependent / independent} reactions / site of carbohydrate production / converts light energy into chemical energy / ignore produces ATP (unless qualified with light dependent stage / ETC) ignore store starch ignore production of oxygen	(1)

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5(a)(ii)	An answer that makes reference to the following points: (transmission) electron microscope (1) - high resolution / allows {the (ultra)structure of the chloroplast to be seen / detailed image} (1)	ignore scanning / electric / electronic allow TEM / electron / EM ignore the image is clear ignore magnification	(2)

Question Number	Answer	Additional guidance	Mark
5(b)(i)	number of (different) species in a {community / this part of the lake / specific area / habitat / ecosystem} (1)	ignore variety of species ignore number of a species	(1)

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5(b)(ii)	An answer that includes the following points: • calculation of N(N-1) (1) • calculation of $\sum n(n-1)$ (1) • calculation of D to 2 d.p. (1)	Species	Number of individuals (n)	(n-1)	n(n-1)	(3)
		A	54	53	2862	
		B	12	11	132	
		C	3	2	6	
		D	31	30	930	
		E	7	6	42	
		F	46	45	2070	
			Total (N) = 153		$\sum n(n-1)$ =6042	
		D= $\frac{23256}{6042}$				
		D = 3.85				
ecf applies						

Question Number	Answer	Additional guidance	Mark
5(b)(iii)	• (the New Zealand pygmy weed has) reduced the {diversity / index / D} of this part of the lake (1) • (due to blocking light) causing death of plants / reduction of {food / habitats} for other organisms (1)	ecf applies from (ii) accept it (out)competed {native species / other plants} / some species {died / left} ecf applies from (ii) e.g. {food source / habitat} for new species	(2)